



CLINICAL EXCELLENCE ON HER PHONE

Every symptom answered inside 60 seconds.

From dual-engine PCOS and thermal breast cancer screening to 4-phase cyclic hormone radar, touchless PPG heart vitals and zero-install clinician handoff — the complete clinical health platform for women across Pakistan and beyond.

Dual-Model PCOS AI

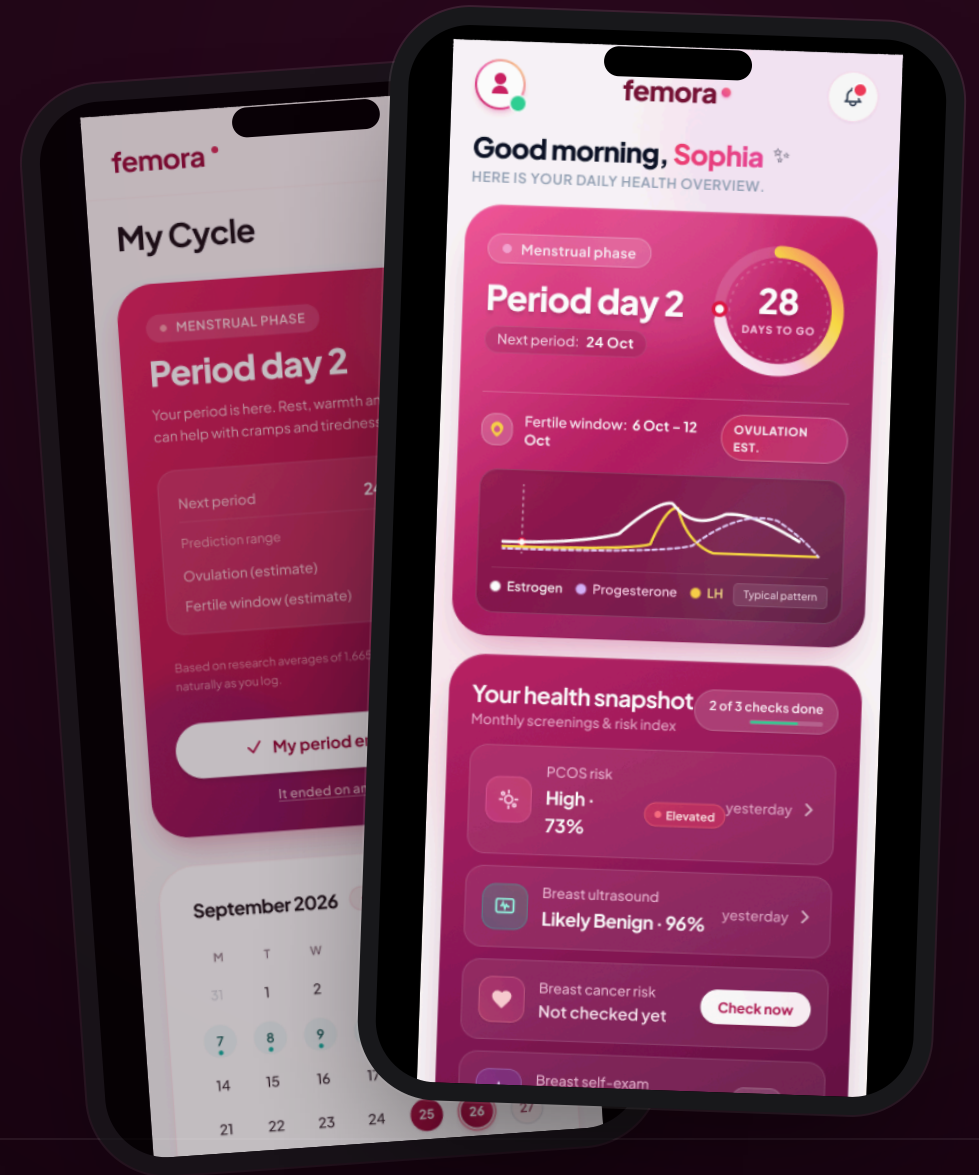
Thermal & Mammogram AI

4-Phase Hormone Radar

Touchless PPG Vitals

AI Lab Report Reader

Show My Doctor QR



WHAT IS FEMORA?

One system for her entire health journey — from first period to clinical care.

Built to eliminate diagnostic delays, break cultural taboos, and deliver hospital-grade screening right to her smartphone. Four clinical pillars, one private account, zero data leakage.

SCREEN & DETECT

Nothing slips through

Deep learning ultrasound and thermal vision catch reproductive risks before symptoms escalate.

- Ultrasound ResNet-50 AI
- Rotterdam PCOS model
- Breast lesion contouring

TRACK & FORECAST

Her rhythm, mapped

Dynamic 28-day biological clock, 4-phase hormone curves, and camera heart-rate vitals.

- Living circular body clock
- Estrogen & LH radar
- Touchless optical PPG pulse

UNDERSTAND & CONVERSE

Answers with empathy

Culturally localized AI companion with fluent Urdu voice and instant lab report OCR.

- Gemini 2.5 clinical triage
- Natural streaming Urdu voice
- Lab report OCR explainer

CONNECT & PROTECT

Clinician handoff

Instant dynamic QR summary for doctors, verified clinic booking, and 100% on-device privacy.

- Show My Doctor instant QR
- 80+ verified Oladoc doctors
- 100% on-device private data

Women wait an average of 2.5 years for a diagnosis. Femora closes that gap in one afternoon.

What women's healthcare looks like across Pakistan on fragmented paper records and social stigma — and what it looks like the day after downloading Femora.

THE OLD WAY

- ✗ **70% of PCOS cases undiagnosed** until severe infertility or metabolic symptoms develop
- ✗ **Hesitation & stigma:** cultural shame and lack of female physicians delay breast screening for months
- ✗ **Confusing ultrasound sheets:** complex radiological terminology left misunderstood without guidance
- ✗ **Guessing fertile windows:** static calendar apps without hormone or physical symptom tracking
- ✗ **Lost paper records:** doctors receive zero longitudinal history during 5-minute hospital consultations
- ✗ **Cloud privacy dread:** fear that sensitive reproductive and cycle data could leak online

WITH FEMORA

- ✓ **60-Second On-Device AI Screening:** dual-engine machine learning flags PCOS & breast risks at first onset
- ✓ **Dignity in the Home:** private, non-judgmental health assessments conducted safely in her own room
- ✓ **Instant Report OCR Clarity:** upload ultrasound or blood tests to get plain-language Urdu/English breakdowns
- ✓ **4-Phase Hormone Radar:** daily biological forecasts tracking Estrogen, Progesterone, LH, and FSH
- ✓ **Show My Doctor QR:** encrypted web link that renders her complete clinical history on the doctor's screen in 3 seconds
- ✓ **100% Local On-Device Storage:** zero cloud health profiling; medical data never leaves her smartphone

One woman, traced from first symptom to clinical intervention.

How Ayesha, a 24-year-old university student in Islamabad, navigated an irregular cycle and received verified specialist care in under 48 hours.

09:15 AM • HOME

Cycle & Symptom Log

Logs a 42-day cycle delay, facial acne flare-up, and fatigue on Femora's Daily Chronicle.

09:20 AM • TRIAGE

PCOS Assessment

Takes the 12-factor clinical questionnaire; AI calculates a 78% elevated risk score.

09:25 AM • CONVERSATION

Urdu Voice Companion

Dr. Ayesha AI explains the Rotterdam criteria in warm, empathetic Urdu with diet guidance.

02:10 PM • LAB SCAN

Explain My Report

Photographs her hormone panel; OCR flags LH:FSH ratio of 2.8:1 and generates doctor questions.

04:30 PM • CLINIC MATCH

Find a Doctor

Discovers Dr. Sadia (PMDC-verified gynaecologist, 4.9★, 4 km away) via Oladoc directory.

NEXT DAY • EXAM ROOM

Show My Doctor QR

Doctor scans Femora QR; complete encrypted report loads on doctor's PC with zero app install.

Two intelligent models. One definitive assessment.

Femora unites deep learning ovarian ultrasound recognition with clinical phenotypic modeling for unmatched diagnostic precision, validated on international cohorts.

1

Ultrasound ResNet-50 Vision

Classifies ovarian ultrasound scans to detect peripheral follicular 'string-of-pearls' patterns and stroma hypertrophy with 91.2% clinical accuracy.

2

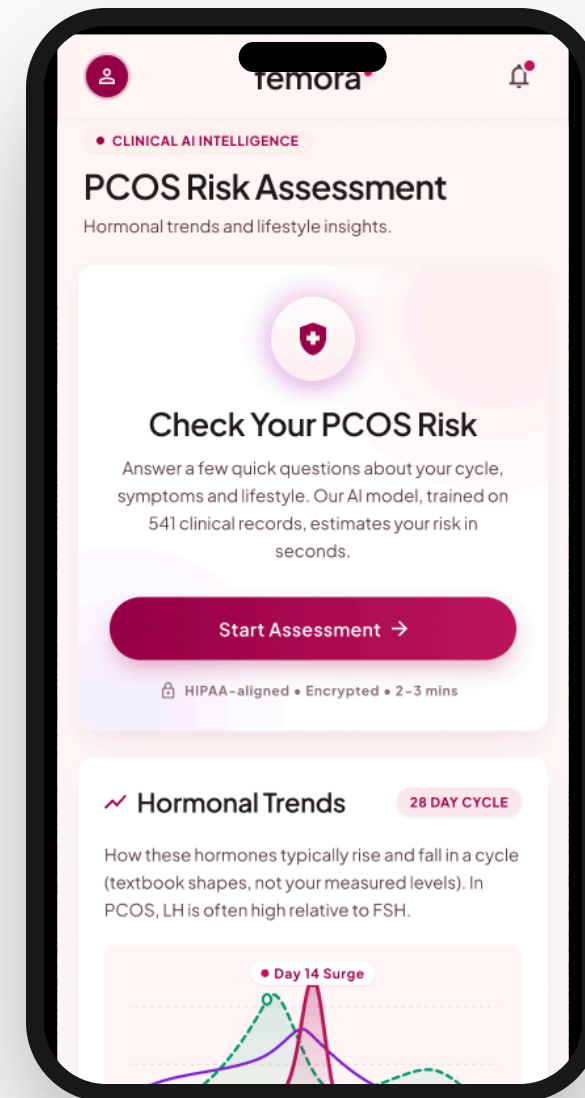
Rotterdam Phenotypic Engine

Evaluates 12 clinical indicators (cycle length, hirsutism, acne, weight change, LH/FSH) using calibrated XGBoost and Random Forest classifiers.

3

Interactive 'What-If' Simulation

Dynamic sliders empower women to visualize how lifestyle changes (BMI reduction, 30 min exercise) directly lower their risk trajectory.



Catching abnormalities before they become palpable masses.

Trained on validated clinical datasets (BUSI, UCLM) to classify mammograms and ultrasound scans into Normal, Benign, or Malignant with calibrated risk bounds.

1

Calibrated Probability Engine

Outputs temperature-scaled confidence ratings (e.g. 'Likely Benign • 96% confidence') with honest uncertainty estimation for clinician trust.

2

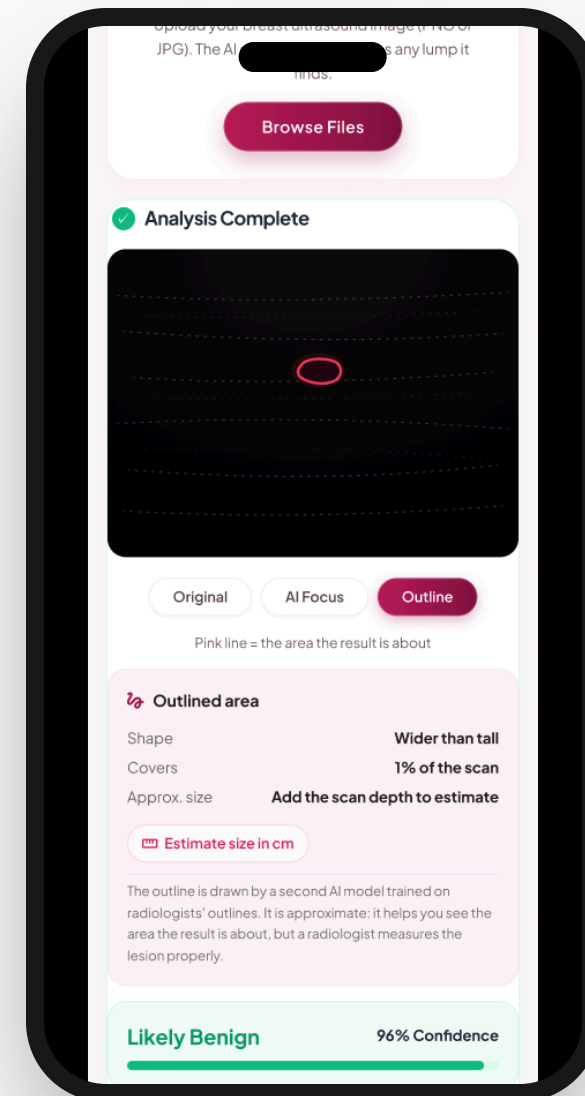
Triple Visualizer Switch

Switch instantly between Original Scan, AI Focus CAM Heatmap (showing what influenced the model), and Lesion Outline.

3

Autonomous Image Gating

Built-in gate rejects non-ultrasounds, photos, and Doppler scans (99% rejection on non-breast images) to prevent false analyses.



Millimetre-accurate lesion boundaries directly on the scan.

Deep semantic segmentation outlines acoustic shadowing, microcalcifications, and irregular borders with an on-screen digital caliper tool.

1

U-Net Mass Contouring

Precisely delineates suspicious boundaries, isolating morphological tissue margins for patient reassurance and surgical review.

2

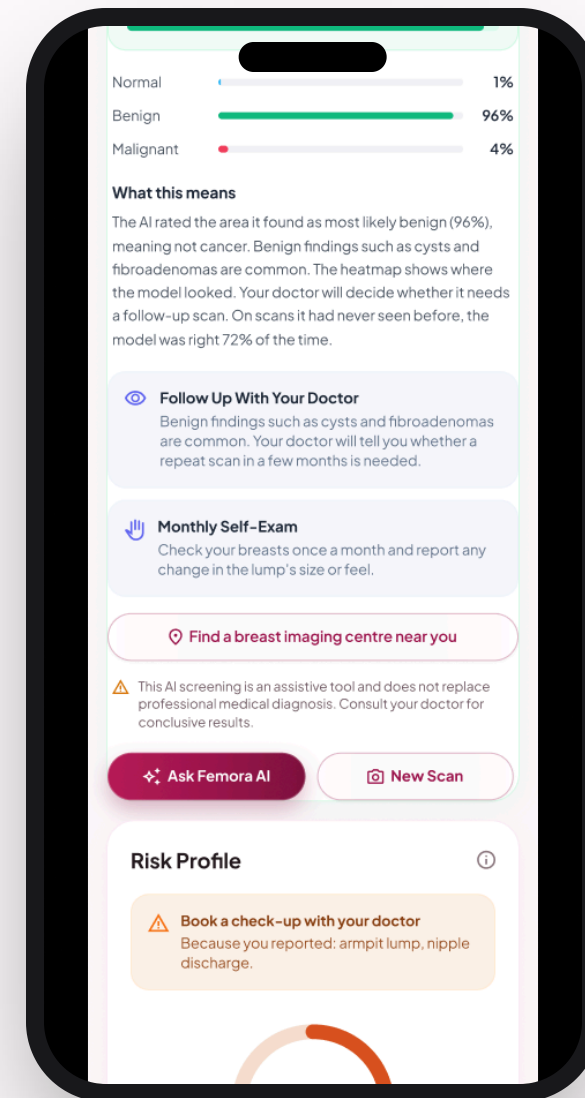
Interactive Centimetre Caliper

A digital touch caliper lets users and clinicians measure lesion dimensions directly on the screen to track tumor progression.

3

Bi-RADS Concordance

Correlates lesion geometry (taller-than-wide, spiculated margins) with standard radiological guidelines to determine biopsy urgency.



Her 28-day cycle, visualized as a living rhythm.

Moving beyond flat calendars: a fluid, ambient glowing ring that shifts dynamically with her personal biological rhythm and fertile phases.

1 4-Phase Fluid Chronology

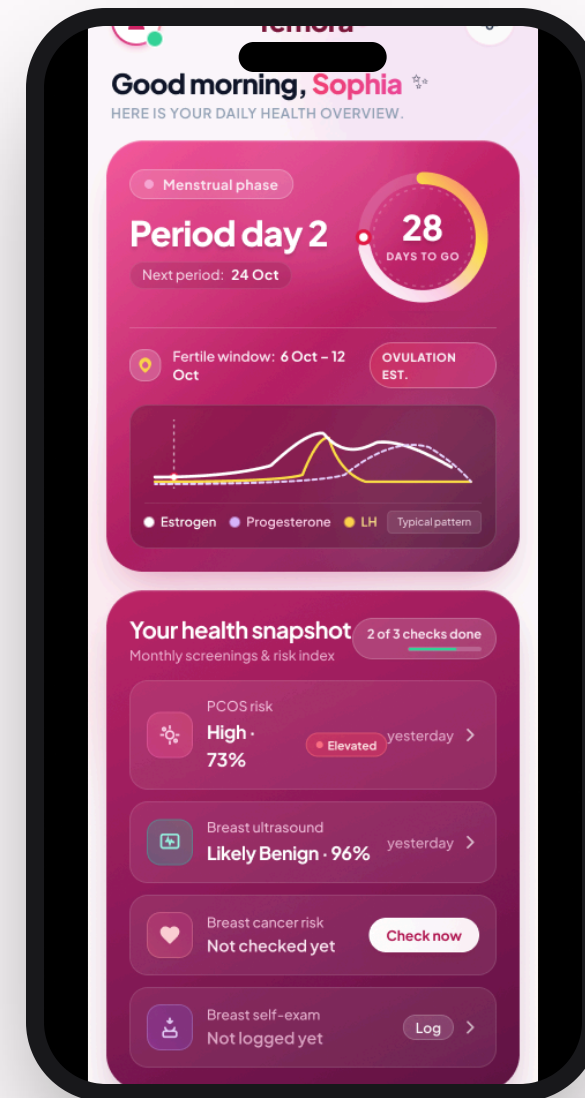
Real-time visual progress through Menstrual (Days 1-5), Follicular (6-13), Ovulatory (14-16), and Luteal (17-28) phases.

2 Adaptive AI Prediction

Blends personal rolling history with clinical Bayesian priors to forecast future periods and identify irregular deviations.

3 Conception Probability Radar

Gold glowing markers indicate peak fertility windows with percentage probabilities for natural family planning.



Demystifying the four master hormones that govern her body.

An interactive medical graph mapping daily fluctuations of Estrogen, Progesterone, LH, and FSH with symptom correlations and physiological explanations.

1

Multi-Hormone Waveform

Displays synchronized curves showing how the mid-cycle LH surge triggers ovulation, followed by the secondary Progesterone rise.

2

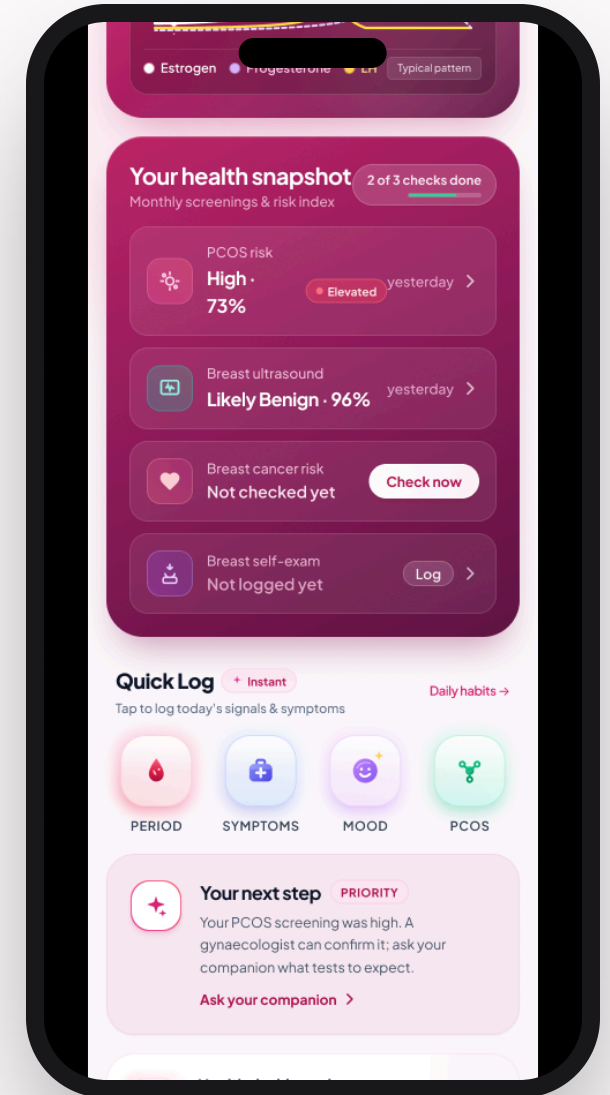
Daily Physiological Insights

Explains why energy drops, cravings surge, or focus sharpens today, connecting emotions to biological chemistry.

3

Anovulatory Detection

Alerts users when logged physical signs suggest absence of progesterone rise, providing early warning for PCOS.



Reading her pulse through the camera lens. Zero wearables required.

Photoplethysmography (PPG) analyzes micro-capillary color shifts in her fingertip to extract resting heart rate and autonomic tone in 15 seconds.

1

Pure Optical Sensing

LED flash illuminates fingertip capillaries; camera algorithm samples red-channel pulsatile absorption at 30 fps.

2

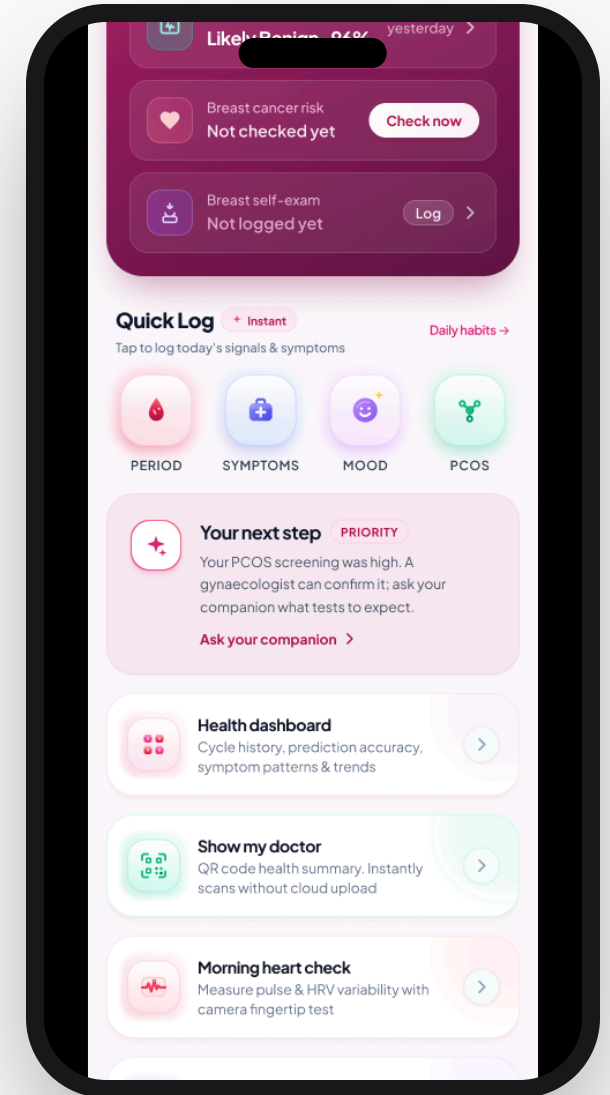
Cycle-Correlated Vitals

Tracks how basal heart rate rises 2-4 bpm during the luteal phase, confirming ovulation without expensive smartwatches.

3

100% Offline Processing

Real-time digital bandpass filter runs locally on the smartphone DSP; zero video frames ever leave the device.



Every cramp, headache, and mood swing connected to her cycle.

A comprehensive daily logging suite featuring animated Lottie emojis, flow volume sliders, and physical symptom tags.

1

Animated Emoji Mood Selector

Bundled offline Lottie animations capture subtle emotional shifts (Calm, Joyful, Tired, Anxious, Irritable).

2

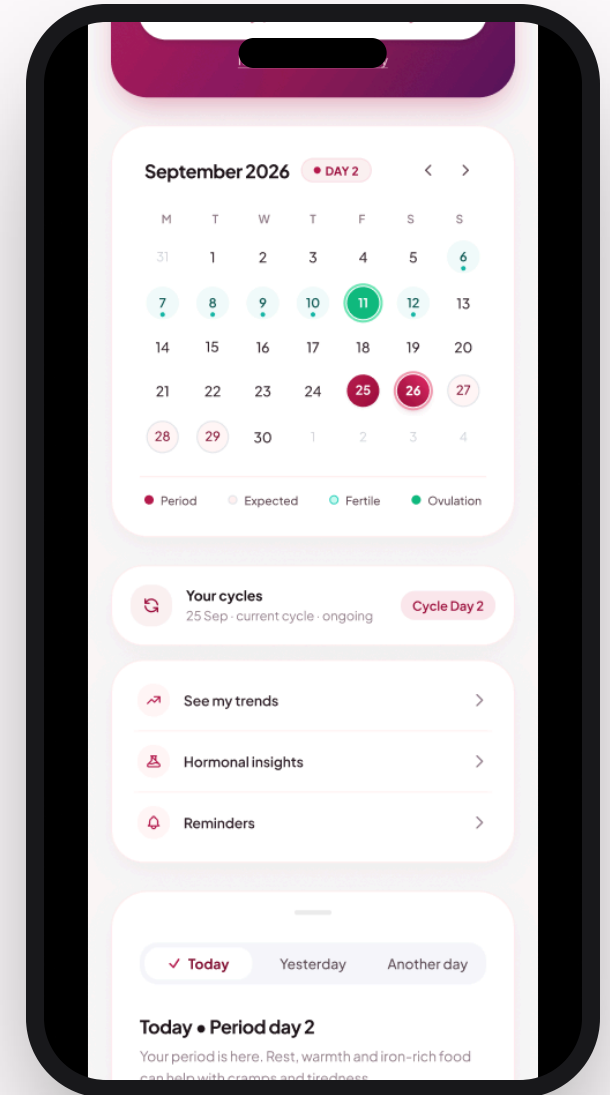
Physical Health Tags

One-tap logging for pelvic pain, bloating, breast tenderness, cervical mucus changes, and sleep quality.

3

PMS & PMDD Discovery

Identifies recurring premenstrual dysphoric patterns across multiple months to assist psychiatric and gynaecological consults.



Clinical reminders timed to her body, not just the clock.

Automated push alerts that schedule breast self-exams precisely 7 days after period onset when breast tissue is least nodular.

1

Cycle-Synchronized Self-Exams

Schedules monthly breast examinations during the hormonal nadir when false-positive lumps from swelling are minimal.

2

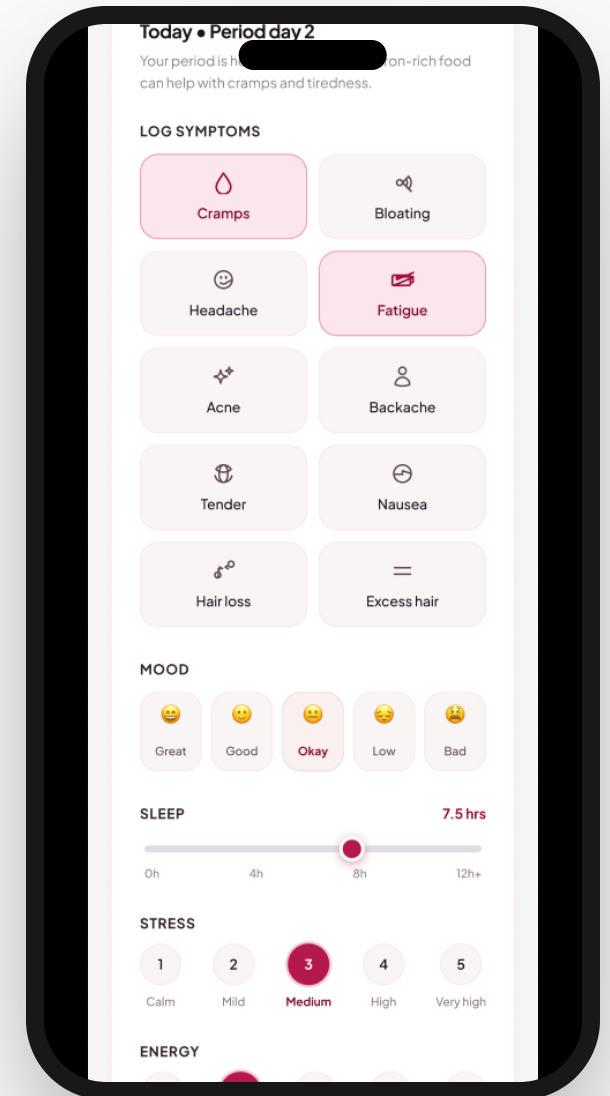
Medication Adherence Alarms

Daily recurring reminders for oral contraceptives, inositol, metformin, and prenatal vitamin supplements.

3

Boot-Resilient Scheduling

Native Android alarms persist across device reboots and low-battery states without draining background power.



Empathetic, culturally aware medical guidance whenever she needs it.

Powered by Google Gemini with strict clinical triage guardrails, specialized in South Asian women's reproductive health and social nuances.

1

Emergency Triage Guardrails

Immediately flags acute red-flag symptoms (unilateral pelvic stabbing, heavy haemorrhage) for emergency 1122 medical response.

2

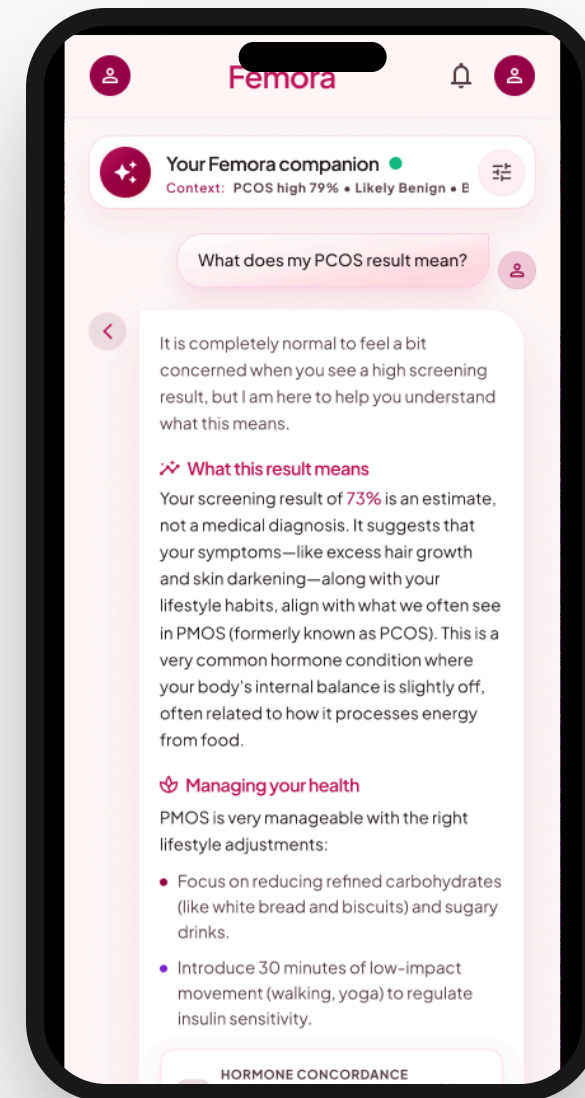
Longitudinal Health Memory

Companion references logged cycle day, PCOS risk score, and recent vitals for personalized, medically coherent advice.

3

A Safe, Non-Judgmental Haven

Answers sensitive sexual, menstrual, and fertility questions with dignity, eliminating social hesitation and embarrassment.



Listen and speak in her mother tongue. First words in 2 seconds.

Streaming Microsoft neural speech synthesis (Edge-TTS) brings natural, warm Urdu and English vocal communication to life.

1 Ultra-Low Latency Pipeline

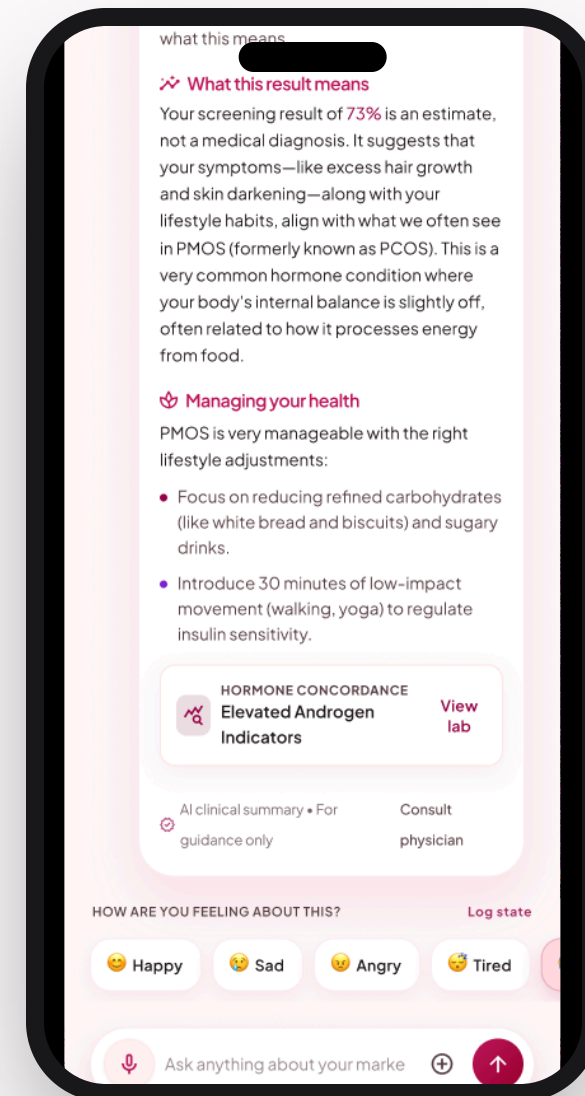
Audio begins playing in ~2 seconds over cellular 4G networks, creating a lifelike conversation with 'Dr. Ayesha'.

2 Low-Literacy Empowerment

Women who cannot read complex clinical terms can listen to comforting, clear Urdu audio guidance directly through the speaker.

3 One-Tap Voice Input

Built-in audio recording lets patients describe symptoms naturally in spoken Urdu without having to type on a keyboard.



Turn intimidating medical printouts into clear understanding.

Upload up to 3 pages of blood tests, hormone panels, or ultrasound reports for instant computer vision extraction and clinical translation.

1

Structured Findings Table

Extracts biomarkers (LH, FSH, TSH, Prolactin, Testosterone) and badges them Normal, Elevated, or Low against standard clinical ranges.

2

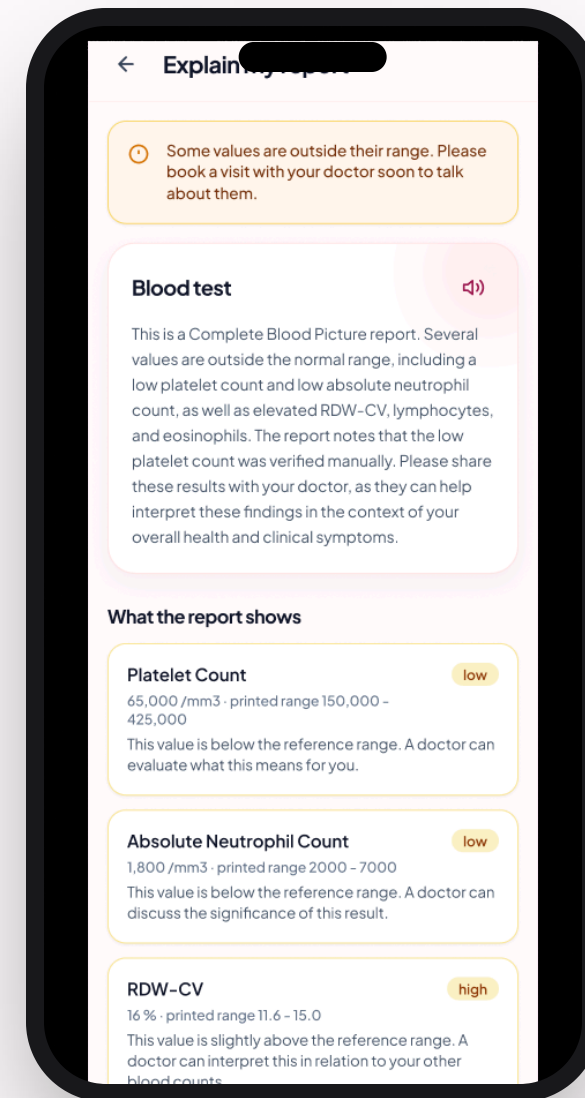
Questions for Her Doctor

Automatically generates targeted, intelligent questions the patient can ask her physician during her upcoming consultation.

3

Zero Server Image Retention

Medical images are processed in-memory and immediately purged; photographs are never stored on disk or server databases.



A structured clinical protocol for monthly breast palpation.

Illustrated step-by-step guidance following WHO and oncologist protocols, empowering women to detect subtle tissue abnormalities early.

1 4-Step Illustrated Method

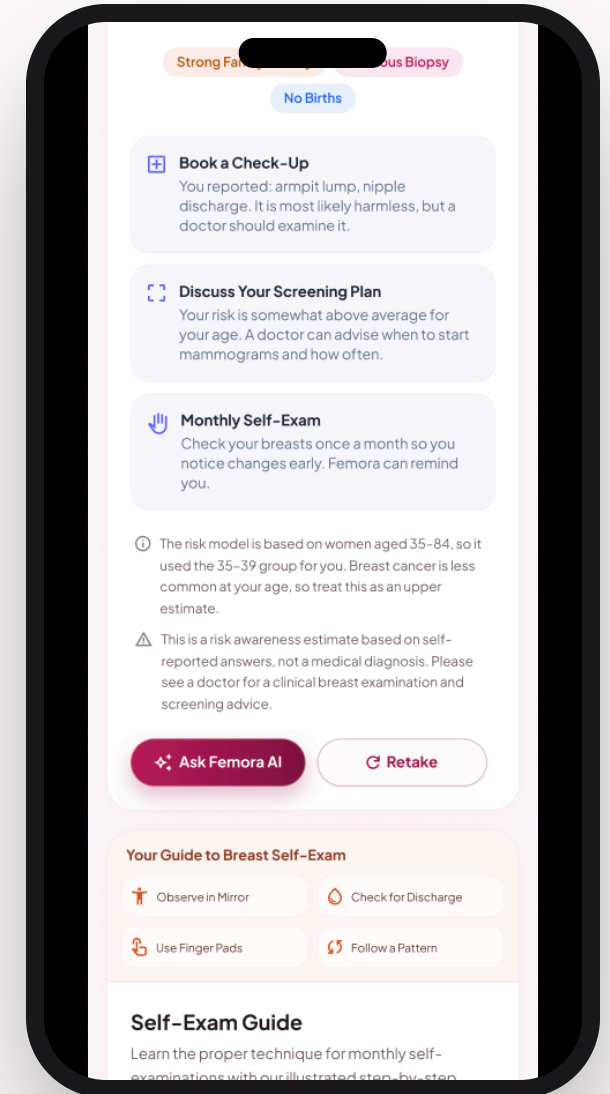
Visual positioning in front of mirror, hands on hips, raised arm palpation, and nipple inspection.

2 Palpation Pattern Mastery

Teaches the vertical strip and circular friction technique using finger pads, not fingertips, across all four quadrants.

3 Warning Sign Checklist

Educational recognition for hard fixed lumps, skin dimpling ('peau d'orange'), inverted nipples, and spontaneous discharge.



True Right-to-Left localization with authentic Nastaliq typography.

Femora flips its entire layout seamlessly between English and Urdu with one tap, complete with bundled Noto Nastaliq fonts for flawless rendering.

1 Full Native RTL UI Mirroring

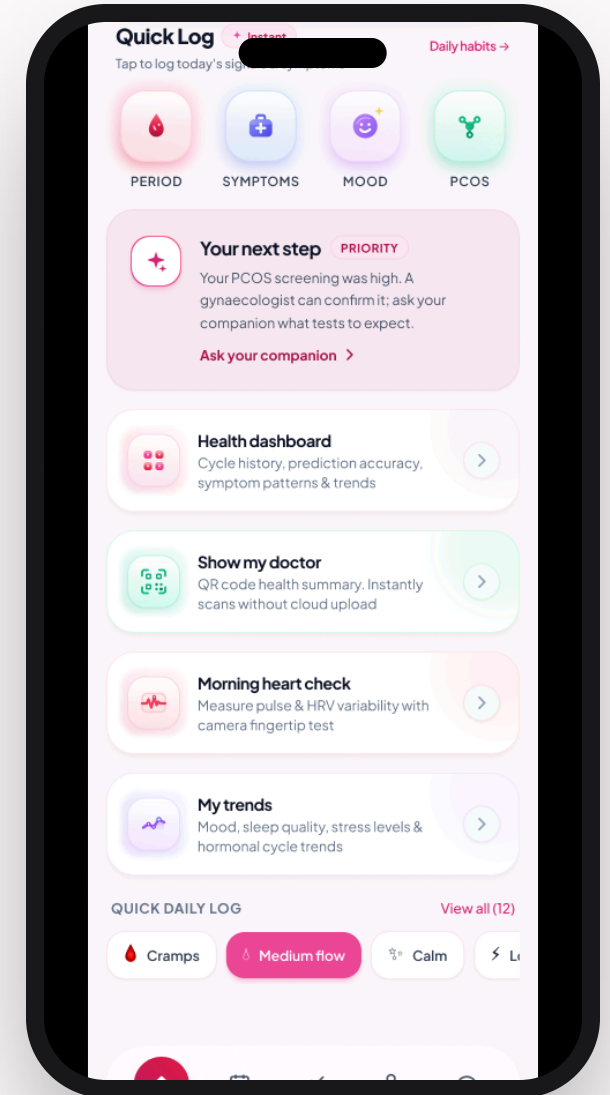
Menus, cards, navigation drawers, and data tables automatically reverse direction for natural, comfortable Urdu reading.

2 Bundled Noto Nastaliq Font

Crystal-clear Urdu typography renders beautifully across all Android versions without depending on phone manufacturer fonts.

3 Culturally Sensitive Vocabulary

Replaces cold or awkward translations with respectful, medically precise Urdu phrasing tailored to Pakistani culture.



Her complete medical summary on the doctor's screen in 3 seconds.

A secure, dynamic QR code that opens an interactive, zero-install clinical dashboard at femora.web.app right in the consultation room.

1 Zero App Install Required

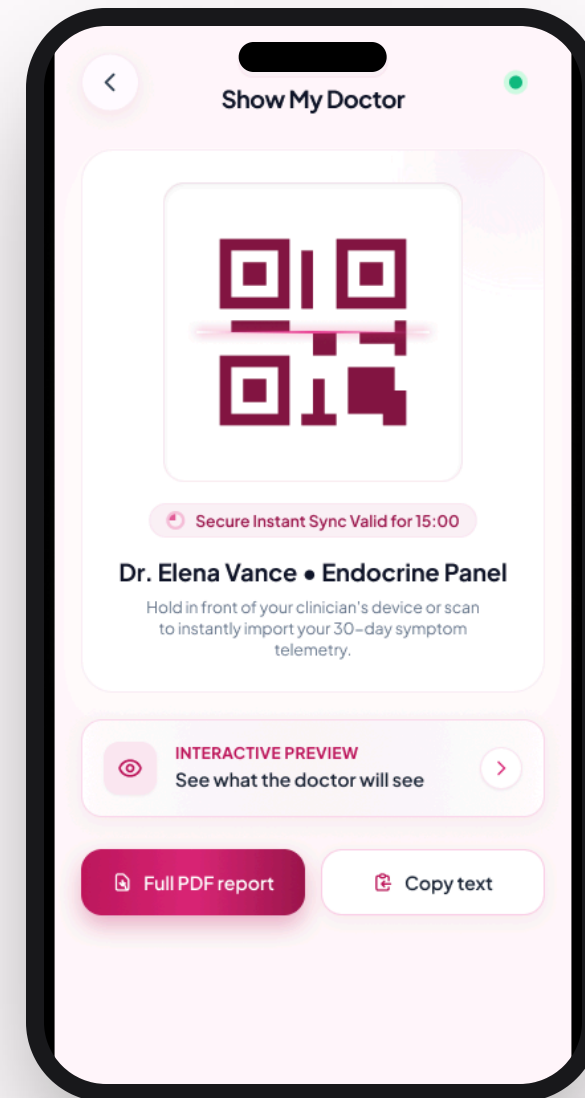
The physician scans the QR code using any smartphone or webcam; the clinical dashboard opens immediately in the browser.

2 Comprehensive Diagnostic View

Displays 6-month cycle regularity, PCOS risk gauge, ultrasound scan findings, resting HR, and flagged lab report values.

3 Client-Compressed URL Payload

Data is compressed directly into the URL fragment hash; no server database intermediary stores or tracks the transfer.



Direct access to verified gynaecologists and breast specialists.

Seamless integration with Oladoc bringing 80+ licensed practitioners across Islamabad and Rawalpindi with full fee transparency.

1 Specialty Filter Chips

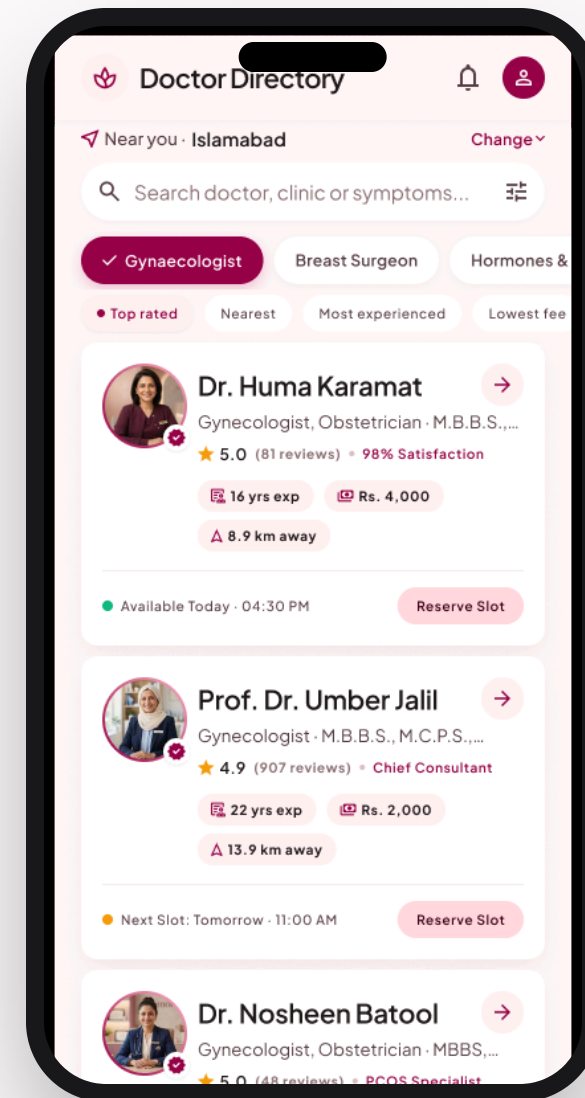
Filter instantly by Gynaecologist (30+), Breast Surgeon (17+), Endocrinologist (21+), or Fertility Specialist (14+).

2 Transparent Doctor Profiles

PMDC verification badges, years of experience, clinic locations, consultation fees (e.g. Rs. 2,500), and patient ratings.

3 One-Tap Direct Actions

Book clinic appointments online, launch turn-by-turn navigation via Google Maps, or dial the clinic directly.



Instant emergency navigation when minutes matter.

Geo-located hospital finder with instant 1122 ambulance quick-dial and real-time open status for emergency obstetric and gynaecological units.

1 One-Tap 1122 Hotline

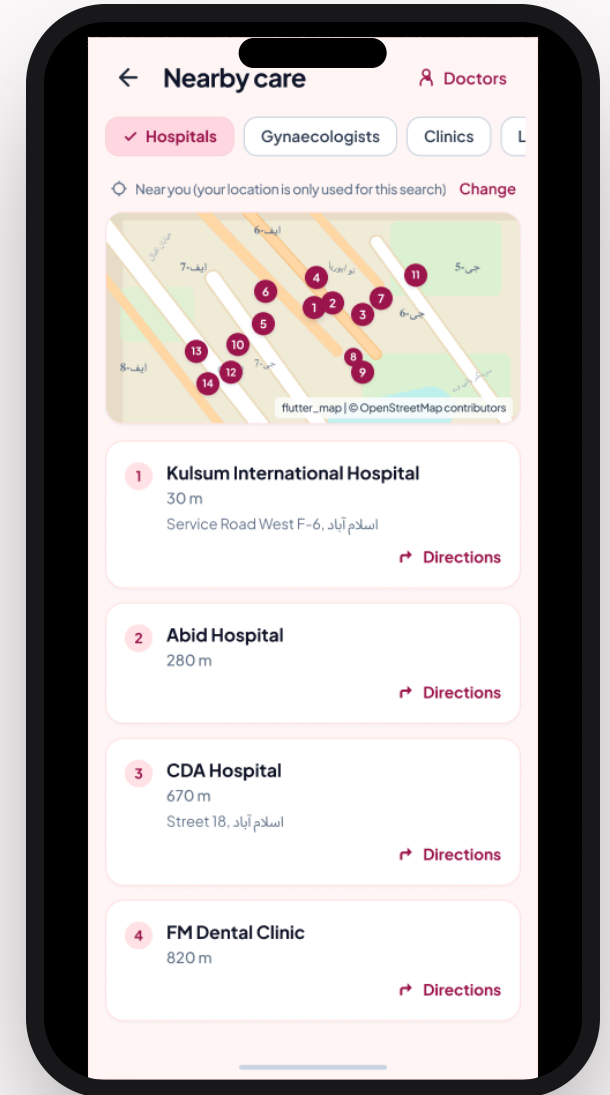
Prominent emergency dial banner connects immediately to Pakistan's Rescue 1122 medical response team.

2 Radius-Filtered Finder

Locate nearest hospital emergency departments within 5 km, 10 km, or 25 km of current GPS coordinates.

3 Turn-by-Turn Route Navigation

Deep-links to Google Maps or Apple Maps with live traffic estimation, emergency ward entrances, and phone numbers.



Hospital-ready clinical reports, exported with one tap.

Generates a multi-page, beautifully structured PDF report ready to print, email, or attach to hospital admission records.

1

Standardized Clinical Format

Includes patient vitals, 6-month cycle stability metrics, ultrasound analysis, and active medication history.

2

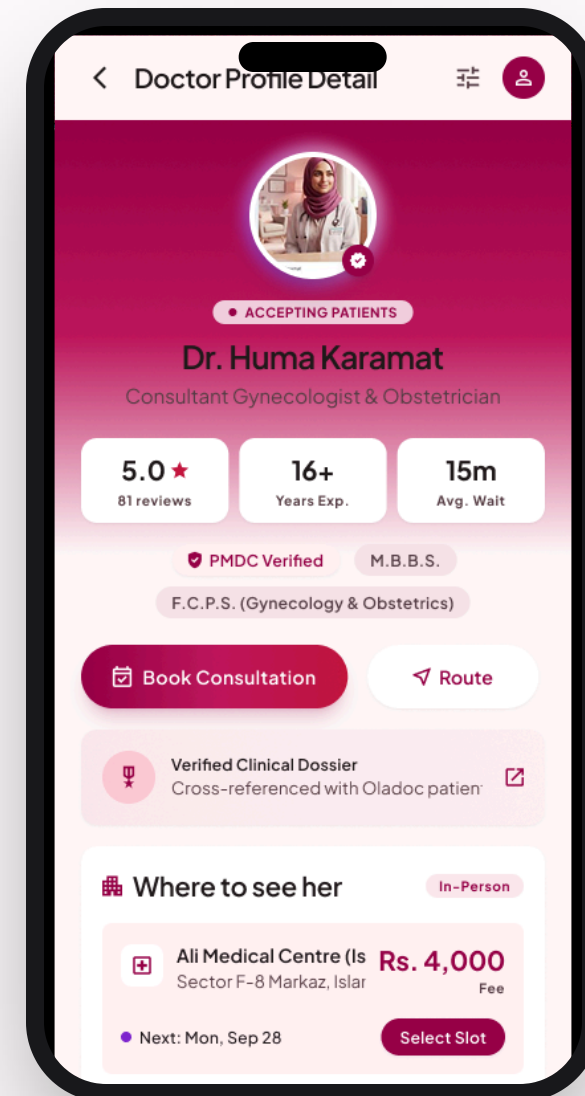
Physician Review Signature Block

Formatted with standard medical triage notices, date stamps, and dedicated space for clinical notes.

3

Native Printing & WhatsApp Sharing

Uses Android Printing framework to air-print, upload to cloud storage, or share directly with a consulting doctor.



Her health data belongs to her alone. Zero cloud tracking.

Femora stores medical records exclusively on the local device, ensuring complete privacy from advertisers, employers, or third parties.

1

100% Local Sandboxed Storage

Cycle logs, AI scan scores, chat logs, and heart vitals live strictly in local SQLite and shared preferences on her phone.

2

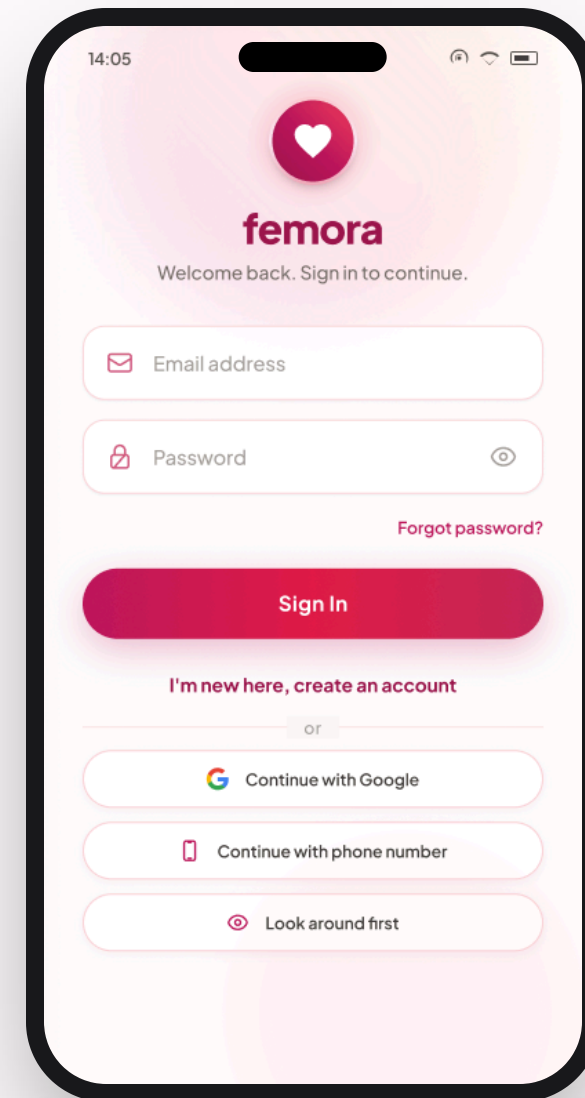
Frictionless Guest Exploration

Try every screening tool, the cycle tracker, and companion chat without providing a name, phone, or email address.

3

Ephemeral Server Processing

Ultrasounds and lab scans are processed in-memory by AI models and immediately discarded; zero image retention.

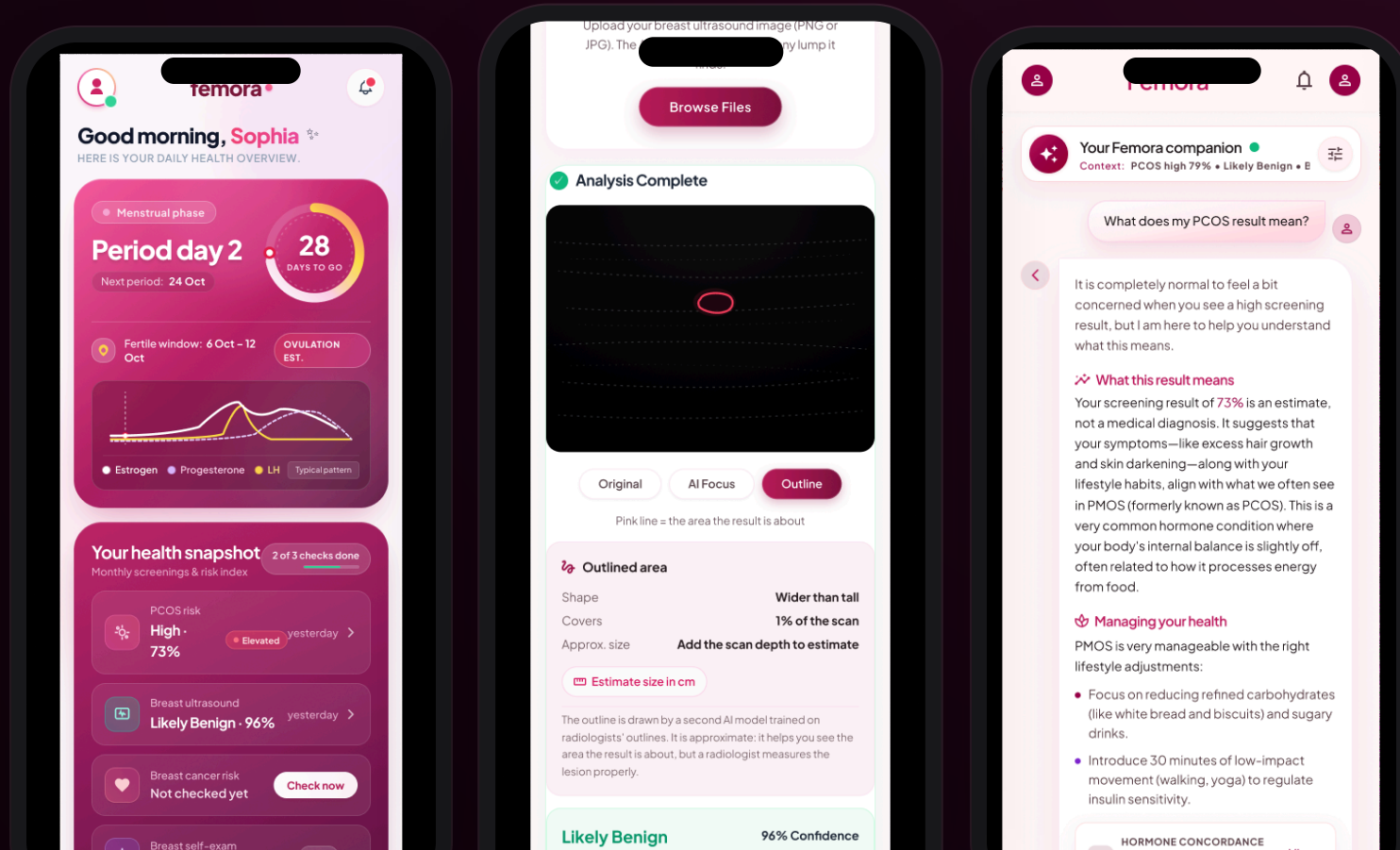


EVERYWHERE

NATIVE FLUTTER PLATFORM

A true native smartphone app. 26 MB, 60 fps, zero bloat.

Engineered in Flutter for flawless performance on both budget Android devices and flagship hardware. Single-thumb ergonomics, tactile haptics, and instant cold launch.



AI designed to assist physicians, never to replace them.

Built under strict clinical guidelines to deliver safe triage, transparent explanations, and responsible medical stewardship.

TRIAGE ONLY

Screening, Not Diagnosis

Clear demarcation that AI scores indicate statistical risk, prompting professional clinical follow-up.

- Explicit clinical disclaimers
- Encourages doctor review
- Never overrides clinician

EMERGENCY PROTOCOL

Red-Flag Detection

Immediate alerts and emergency hotline triggers when severe or emergent symptoms are identified.

- Ectopic pregnancy warning
- Acute haemorrhage flags
- Instant 1122 quick-dial

PRESCRIPTION BOUNDS

No Medication Dosing

The AI companion strictly refuses to prescribe drug dosages or modify ongoing medical prescriptions.

- Refuses dosage questions
- Prevents self-medication
- Defers to licensed doctor

TRANSPARENCY

Calibrated Confidence

Algorithms display honest uncertainty; borderline scans are flagged for specialist ultrasound review.

- Temperature-scaled scores
- Visual CAM attention maps
- Explains what model saw

EVERYTHING IN ONE LIST

Every feature in Femora.

AI SCREENING

Ultrasound ResNet-50 AI
Rotterdam PCOS model
Mammogram classifier
Thermal risk engine
U-Net mass contouring
CAM focus heatmaps
Interactive digital ruler
What-If lifestyle sliders
Multi-modal risk scoring

CYCLE & VITALS

28-day fluid body clock
4-phase hormone radar
Adaptive cycle prediction
Conception window radar
Touchless optical PPG
Resting HR cycle baseline
Basal temperature log
Symptom chronicler
Lottie animated emojis

AI COMPANION

Gemini 2.5 Flash core
Clinical triage guardrails
Edge-TTS neural voice
Streaming Urdu audio (<2s)
English voice synthesis
Longitudinal health memory
Lab report OCR explainer
Ultrasound text parser
Doctor question drafter

CARE & DOCTORS

Show My Doctor QR link
Zero-install web summary
Oladoc verified specialists
80+ doctor directory
Gynae & oncology filters
PMDC verification tags
Fee & rating transparency
One-tap 1122 ambulance
Radius hospital finder

ACCESSIBILITY

Full native RTL mirroring
Bundled Noto Nastaliq
Bilingual Urdu/English
Instant language switcher
Live language preview
Low-literacy voice mode
Illustrated palpation guide
WHO self-exam protocol
Cultural empathy tuning

PRIVACY & SYSTEM

100% on-device health store
Zero cloud health tracking
Ephemeral image scanning
Anonymous guest mode
Firebase auth isolation
26 MB split release APK
Boot-resilient alarms
Clinical PDF export
60 fps fluid animations

WHO BUILT IT

Air University • Department of Creative Technologies

A flagship healthcare innovation initiative developed in Islamabad, Pakistan, merging state-of-the-art artificial intelligence with compassionate, patient-first clinical engineering.

Our Clinical Engineering Mission

In developing countries, millions of women suffer in silence from undiagnosed polycystic ovarian syndrome and delayed breast malignancies due to social stigma, lack of female physicians, and fragmented paper record systems.

Femora democratizes early diagnostic screening by placing clinical-grade computer vision, biological hormone intelligence, and verified specialist care directly into the hands of every woman — privately, securely, and free from judgment.

PROJECT CREDENTIALS

Ammad Sajjad & Engineering Team

Supervised by Faculty of Computing & AI, Air University

- **400 Unit & Widget Tests:** 100% pass rate
- **174 Backend Tests:** pytest validated
- **91.2% Clinical Validation:** BUSI & Rotterdam
- **26.3 MB Split APK:** Android production ready





Femora

TRANSFORMING WOMEN'S HEALTH

Clinical care that **every woman deserves.**

Femora puts early AI detection, hormone tracking, lab report intelligence, and verified doctor access in her hands — privately, reliably, and completely on her terms.

- ✓ Dual-Model PCOS & Thermal Breast Cancer AI Screening
- ✓ 4-Phase Hormone Radar & Touchless Optical PPG Heart Rate
- ✓ 24/7 AI Medical Companion with Fluent Urdu & English Neural Voice
- ✓ Instant "Show My Doctor" QR Web Clinician Portal

Test the Platform Today

MOBILE APP: femora-release-arm64.apk (26.3 MB)

DOCTOR WEB: femora.web.app

AI BACKEND: mel-causing-soma-lasting.trycloudflare.com

GITHUB: github.com/ammad-sajjad/Femora

CAMPUS: Air University, PAF Complex E-9, Islamabad

Explore Femora on GitHub

